EE2080 : Lecture 11 : 22 August 2025

RECAP: For a time homogeneous DTMC $\{X_n, n \geq 0\}$ on the state space S.

- Transient, Null-recurrent & Positive Recurrent States
- Communicating classes & Class property theorem
- Irreducible DTMC, positive recurrence and a unique stationary distribution

PLAN: Examples

Irreducibility and Periodicity of a TH DTMC

Proof of Class Property Theorem.

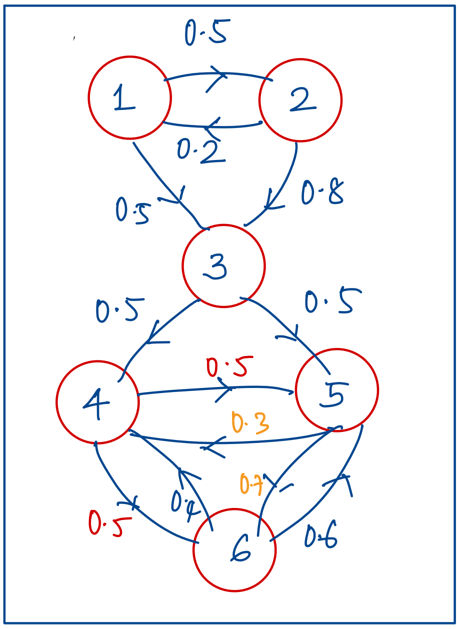
Result about open/closed communicating classes

Announcement!

Programming Assignment

- 3 qns.
  - i) From Q1 in PS1 (5)
  - ii) Gambler's Ruin (10)
  - iii) DTMC (15)

Example: Consider the TH DTMC with the following transition probability diagram.



1) Transition Probability Matrix P

2) Assuming $\pi(0) = \text{unif}\{1,2,3,4,5,6\}$, find $\pi(1), \pi(2)$

3) The first passage time distributions : $f_{ii}^{(n)}$, for all $i \in S$.

   and find f_{ii}, $\forall i \in \{1,2,\ldots,6\}$.

4) Classify the states as transient & recurrent based on the values of f_{ii} above

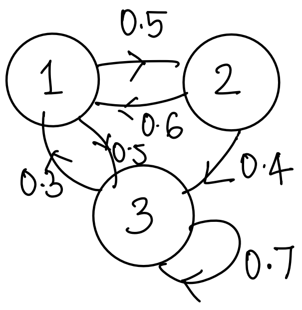
5) Reachability matrix and identify the communicating classes

6) If there are recurrent states, find their mean recurrence time ν_{jj}

7) Based on ν_{jj}, classify the recurrent states as positive & null recurrent.

8) Find the stationary distribution π s.t $\pi = \pi P$.

Example:



1) Find the stationary distribution π s.t $\pi = \pi P$, $\sum_{i=1}^{3} \pi_i = 1$

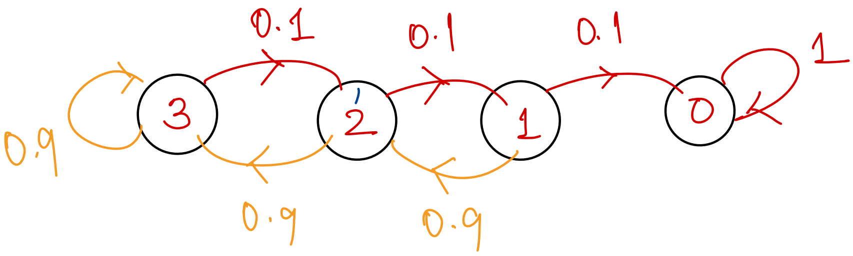
2) Find f_{11}, f_{22}, f_{33}.

3) Are the states transient & recurrent?

4) Find $\nu_{11}, \nu_{22}, \nu_{33}$. 5) Are they related to π?

Goldmann Sachs Interview Question @ IISc (2017-18 Placement Season)

Imagine that you are playing a game that involves tossing a coin of bias 0.9 repeatedly and independently, with 3 lives at the start of the game. At each round, if the coin toss results in a tail, you lose one life. However, if the coin toss results in a head, you gain back your lost lives one at a time. If you have all 3 lives and the coin toss results in a head, nothing changes and you continue tossing. You are allowed to play until you lose all your lives. What is the expected number of times you will toss the coin ?



n_i – the average number of tosses to go from i to 0 $\quad i \in \{1,2,3\}$.

$n_0 = 0$

$n_1 = 0.1(1+n_0) + 0.9*(1+n_2)$

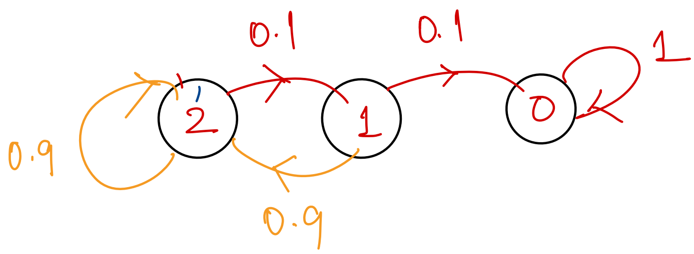
$n_2 = 0.1(1+n_1) + 0.9*(1+n_3)$.

$n_3 = 0.1(1+n_2) + 0.9*(1+n_3)$

Required to find n_3.

$n_3 = \mathbb{E}\left[T_0^{(1)} \middle| X_0 = 3\right]$

What is $f_{30}^{(n)}$?



$$f_{20}^{(n)} = \begin{cases} 0.1^2, & n=2 \\ & \end{cases}$$

EE 2080: Lecture 12: 25 August 2025

PLAN: 1) Periodicity of a DTMC
2) Proofs for "Period, transience, recurrence are class properties"
3) Processes with stationary and independent increment properties.

Irreducibility and Periodicity

A time homogeneous DTMC $\{X_n, n \geq 0\}$ is said to be irreducible if the entire state space S constitutes a single communicating class.

i.e $\forall\, i,j \in S$, $\exists\, n \in \mathbb{N}$ s.t $P_{i,j}^{n} > 0$.

Period: Consider a TH DTMC $\{X_n, n \geq 0\}$ on a state space S with TPM P.

For any $j \in S$, let $R(j) = \{n \in \mathbb{N} \text{ s.t } P_{jj}^{(n)} > 0\}$ denote the set of return times to j.

Defn: The period of a state $j \in S$, denoted $d(j)$ is defined as the greatest common divisor of the set of return times, i.e.,

$$d(j) = \gcd \{n \in \mathbb{N} \text{ s.t } P_{jj}^{(n)} > 0\}$$

A state j is called aperiodic if $d(j) = 1$, and periodic if $d(j) > 1$.

Aperiodic Markov Chain

A time homogeneous DTMC is said to be aperiodic if
i) it is irreducible
ii) every state is aperiodic.

Example: $P_1 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ $\quad P_2 = \begin{bmatrix} 0 & 0.2 & 0 & 0.8 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.9 & 0 & 0.1 \\ 0.7 & 0 & 0.3 & 0 \end{bmatrix}$

Identify whether the DTMC is periodic/aperiodic.
If periodic, what are the periods?

Qn: What if $P = \begin{bmatrix} 0.1 & 0.9 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$?

What do you gather about the periodicity of DTMCs with self loops?

Class Properties

1) Period is a class property, i.e., if $i \leftrightarrow j$, $d(i) = d(j)$

Proof: $i \leftrightarrow j \Rightarrow \exists\, m, n \in \mathbb{N}$ s.t $P_{i,j}^{m} > 0$ and $P_{j,i}^{n} > 0$

$\Rightarrow$ Let $r \in R(i)$, i.e $P_{i,i}^{r} > 0$

$$P_{j,j}^{n+m} \geq P_{j,i}^{n} \cdot P_{i,j}^{m} > 0, \quad P_{j,j}^{n+r+m} \geq P_{j,i}^{n} P_{i,i}^{r} P_{i,j}^{m} > 0.$$

$\therefore\ n+m \in R(j)$ and $n+r+m \in R(j)$

$\Rightarrow d(j) \mid n+m$ and $d(j) \mid n+m+r$

$\Rightarrow d(j) \mid r$

Thus if $r \in R(i)$ and $d(j) \mid r \Rightarrow d(j)$ is a common divisor of $R(i)$

$\Rightarrow d(j) \leq d(i)$ as $d(i)$ is the greatest common divisor of $R(i)$.

Switching the roles of i & j, we can show that $d(i) \leq d(j)$

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2) Transience and Recurrence are class properties, i.e., states within a communicating class are all transient or all recurrent

– Recurrence

Suppose i is recurrent and $i \leftrightarrow j$

Then, $\exists\, m, n \in \mathbb{N}$ s.t $P_{i,j}^{m} > 0$ and $P_{j,i}^{n} > 0$

i recurrent $\Rightarrow \mathbb{E}[M_i \mid X_0 = i] = \sum_{s \in \mathbb{N}} P_{i,i}^{s} = +\infty$ (expected number of returns to state i is infinite)

$$\mathbb{E}[M_j \mid X_0 = j] = \sum_{s \in \mathbb{N}} P_{j,j}^{s}$$

$$= \geq P_{j,j}^{m+n+1} + P_{j,j}^{m+n+2} + \cdots$$

$$= \sum_{s \in \mathbb{N}} P_{j,j}^{m+n+s}$$

$$\geq \sum_{s \in \mathbb{N}} P_{j,i}^{n} P_{i,i}^{s} \cdot P_{i,j}^{m} = P_{j,i}^{n} P_{i,j}^{m} \underbrace{\sum_{s \in \mathbb{N}} P_{i,i}^{s}}_{\infty} = \underline{\underline{+\infty}}$$

- Transience

Suppose i is transient and $i \leftrightarrow j$

Then, $\exists\, m, n \in \mathbb{N}$ s.t $P_{i,j}^{m} > 0$ and $P_{j,i}^{n} > 0$

i recurrent $\Rightarrow \mathbb{E}[M_i \mid X_0 = i] = \sum_{s \in \mathbb{N}} P_{i,i}^{s} < \infty$ (expected number of returns to state i is finite)

$$\mathbb{E}[M_j \mid X_0 = j] = \sum_{s \in \mathbb{N}} P_{j,j}^{s}$$

$$= \frac{1}{P_{i,j}^{m} P_{j,i}^{n}} \sum_{s \in \mathbb{N}} P_{i,j}^{m} P_{j,j}^{s} P_{j,i}^{n}$$

$$\leq \frac{1}{P_{i,j}^{m} P_{j,i}^{n}} \sum_{s \in \mathbb{N}} P_{i,i}^{m+n+s} \leq \underbrace{\frac{1}{P_{i,j}^{m} P_{j,i}^{n}} \sum_{k \in \mathbb{N}} P_{i,i}^{k}}_{< \infty}$$

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3) Positive Recurrence and Null Recurrence are class properties.

Result about open/Closed communicating class

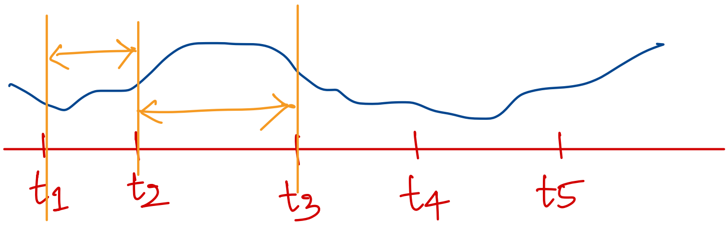
1) If C is an open communicating class, then every state within C is transient.

2) If C is a finite closed Communicating class, then every state within C is positive recurrent.

2 $\Rightarrow$ An irreducible DTMC on a finite state space is positive recurrent.

Stationary and Independent Increments

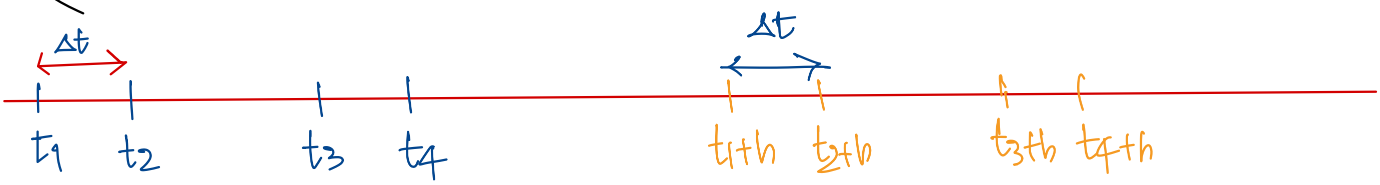
Let $\{X_t, t \in T\}$ be a random process. We say that $X(t)$ has independent increments if $\forall t_1, t_2, \ldots, t_n \in T$ s.t $t_1 < t_2 < t_3 < \ldots < t_n$, $X_{t_2} - X_{t_1}, X_{t_3} - X_{t_2}, \ldots, X_{t_n} - X_{t_{n-1}}$ are independent.



What happens in one interval is independent of what happens in another non-overlapping interval.

A random process $\{X_t : t \geq 0\}$ has stationary increments if $\forall t_1, t_2 \ldots, t_n \in T$ s.t $0 \leq t_1 < t_2 < \cdots < t_n$, and $\forall h > 0$,

$$\left(X_{t_2} - X_{t_1}, X_{t_3} - X_{t_2}, \ldots X_{t_n} - X_{t_{n-1}}\right) \overset{d}{=} \left(X_{t_2+h} - X_{t_1+h}, \ldots X_{t_n+h} - X_{t_{n-1}+h}\right)$$



In particular $X_{t_1+h} - X_{t_1}$ and $X_{t_2+h} - X_{t_2}$ will have the same distribution.

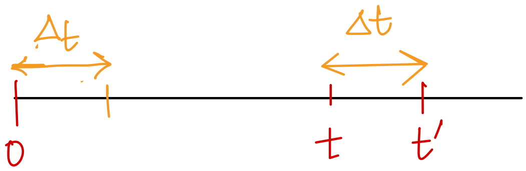
The distribution of the increments depend only on the span of the observation and not on the actual starting time of observation.

Counting Process: A counting process $\{N_t, t \geq 0\}$ has stationary increments if

$N(t') - N(t) \overset{d}{=} N(t'-t)$ for every $t' > t > 0$.

$N(t') - N(t) = $ # arrivals in the interval $[t, t']$

$N(t'-t) = $ Number of arrivals in $(0, t'-t]$



Example: Consider the iid Bernoulli process $\{X_n : n \in \mathbb{N}\}$, $X_n \overset{iid}{\sim} Ber(p)$. (Random Walk)

Counting process: $\{S_n, n \geq 0\}$, where $S_0 = 0$ and $S_n = \sum_{i=1}^{n} X_i$

- Show that $\{S_n\}_{n=0}^{\infty}$ has the stationary & independent increment properties.

Remark: Poisson processes have both the stationary and independent increment properties.